

AIM:- Implement the RSA algorithm for public-key encryption and decryption, and explore its properties and security considerations.

CLI

CODE:-

```
def gcd(a, b):
    while b:
        a, b = b, a % b
    return a

def mod_inverse(e, phi):
    for d in range(1, phi):
        if (d * e) % phi == 1:
            return d
    return None

p = 61
q = 53

n = p * q
phi = (p - 1) * (q - 1)

e = 17

while gcd(e, phi) != 1:
    e += 2

d = mod_inverse(e, phi)

print("RSA ENCRYPTION & DECRYPTION")
print("Public Key :", (e, n))
print("Private Key:", (d, n))

while True:
    print("\n1. Encrypt Message")
    print("2. Decrypt Message")
    print("3. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        message = input("Enter message to encrypt: ")

        encrypted = [pow(ord(ch), e, n) for ch in message]
```

```
print("Encrypted Message:")
print(encrypted)

elif choice == "2":
    data = input("Enter encrypted numbers: ")

    try:
        data = (data.replace("[", "")
                .replace("]", "")
                .replace("(", "")
                .replace(")", "")
                .replace(",", " "))

        encrypted = list(map(int, data.split()))

        decrypted = "".join(chr(pow(num, d, n)) for num in encrypted)

        print("Decrypted Message:")
        print(decrypted)

    except ValueError:
        print("Invalid encrypted data!")

elif choice == "3":
    print("Program Ended.")
    break

else:
    print("Invalid Choice!")
```

OUTPUT:-

```

IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Public/cli_rsa.py =====
RSA ENCRYPTION & DECRYPTION
Public Key : (17, 3233)
Private Key: (2753, 3233)

1. Encrypt Message
2. Decrypt Message
3. Exit
Enter your choice: 1
Enter message to encrypt: Abhishek Singh
Encrypted Message:
[2790, 2570, 2170, 3179, 1230, 2170, 1313, 690, 1992, 2680, 3179, 2235, 2923, 2170]

1. Encrypt Message
2. Decrypt Message
3. Exit
Enter your choice: 2
Enter encrypted numbers: 2790, 2570, 2170, 3179, 1230, 2170, 1313, 690, 1992, 2680, 3179, 2235, 2923, 2170
Decrypted Message:
Abhishek Singh

1. Encrypt Message
2. Decrypt Message
3. Exit
Enter your choice: 3
Program Ended.
>>>

```

GUI**CODE:-**

```

from tkinter import *

def gcd(a, b):
    while b:
        a, b = b, a % b
    return a

def mod_inverse(e, phi):
    for d in range(1, phi):
        if (d * e) % phi == 1:
            return d
    return None

p = 61
q = 53

n = p * q
phi = (p - 1) * (q - 1)

```

```

e = 17

while gcd(e, phi) != 1:
    e += 2

d = mod_inverse(e, phi)

def encrypt():
    message = entry.get()

    encrypted = [pow(ord(ch), e, n) for ch in message]

    output.delete(0, END)
    output.insert(0, str(encrypted))

def decrypt():
    data = output.get()

    try:
        data = (data.replace("[", "")
                .replace("]", "")
                .replace("(", "")
                .replace(")", "")
                .replace(",", " "))

        encrypted = list(map(int, data.split()))

        decrypted = ''.join(chr(pow(num, d, n)) for num in encrypted)

        result.config(text="Decrypted Message: " + decrypted)

    except:
        result.config(text="Invalid Encrypted Data")

def clear():
    entry.delete(0, END)
    output.delete(0, END)
    result.config(text="")

root = Tk()
root.title("RSA Encryption & Decryption")
root.geometry("550x320")
root.resizable(False, False)

Label(root,

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text="RSA Encryption & Decryption",
font=("Arial", 16, "bold")).pack(pady=10)

Label(root, text="Enter Message").pack()

entry = Entry(root, width=55, font=("Arial", 11))
entry.pack(pady=5)

Button(root,
        text="Encrypt",
        width=15,
        command=encrypt).pack(pady=5)

Label(root, text="Encrypted Message").pack()

output = Entry(root, width=55, font=("Arial", 11))
output.pack(pady=5)

Button(root,
        text="Decrypt",
        width=15,
        command=decrypt).pack(pady=5)

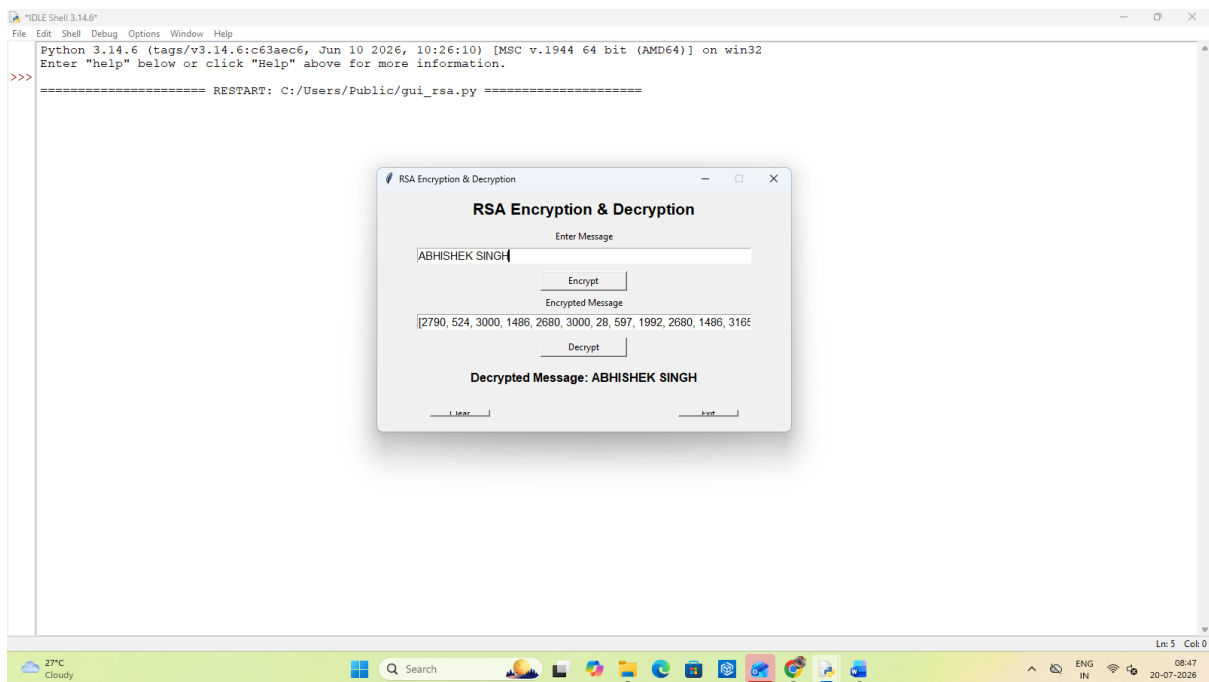
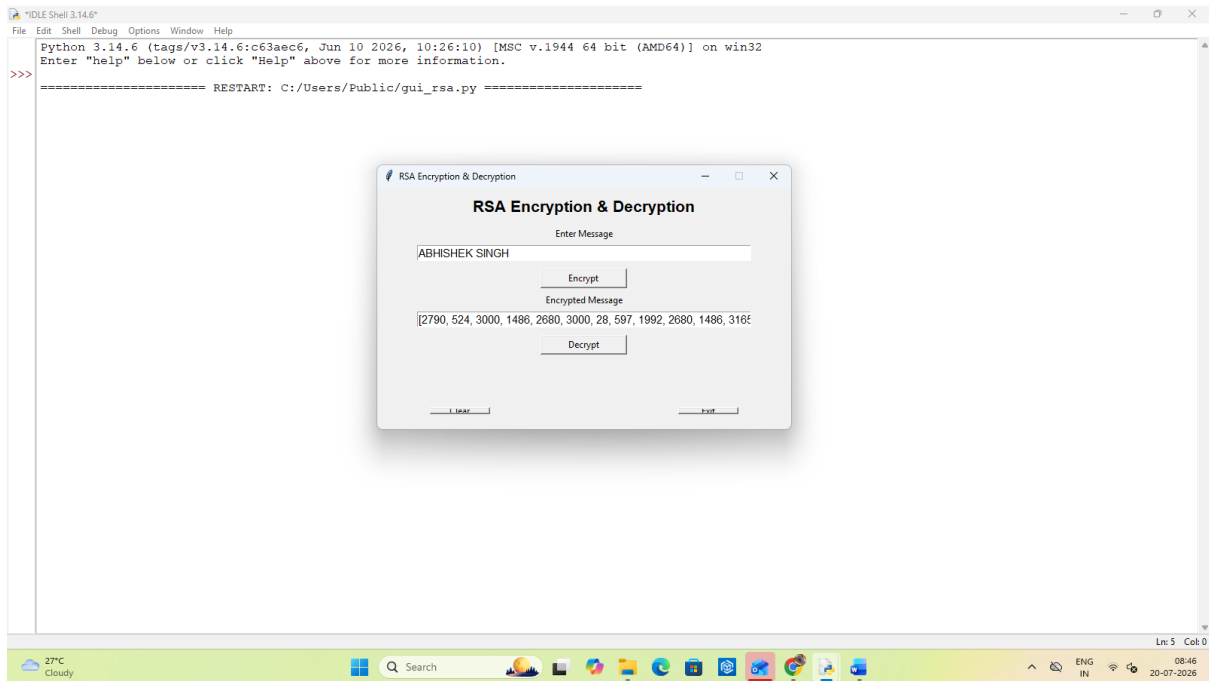
result = Label(root,
                text="",
                font=("Arial", 12, "bold"))
result.pack(pady=10)

Button(root,
        text="Clear",
        width=10,
        command=clear).pack(side=LEFT, padx=70, pady=20)

Button(root,
        text="Exit",
        width=10,
        command=root.destroy).pack(side=RIGHT, padx=70, pady=20)

root.mainloop()
```

OUTPUT:-



ABHISHEK DINESH SINGH
T118